

The Evolution of Search

HONORS THESIS DEFENSE

Comparing Traditional Retrieval to Modern AI-Enhanced Generative
Systems

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WHY SEARCH ENGINES MATTER

The Gateway to Knowledge

Information Access: Search engines are the primary lens through which users research medical, academic, global events, etc.

The "10 Blue Links" Era

Users manually compare sources and synthesize their own conclusions from ranked document lists.

The Generative Shift

AI systems synthesize information into direct answers, reducing user effort but introducing new quality risks.



Research Question: Is the generative format actually surfacing better information, or is it just more persuasive?

FROM RETRIEVAL TO GENERATION

The Paradigm Shift

- **Traditional:** Retrieve and rank documents
- **Modern AI:** Synthesizing direct answers using Large Language Models.
- **Goal:** Shift from "Finding Information" to "Answering Questions."
- **Impact:** Fundamental change in user experience and cognitive load.



TRADITIONAL SEARCH FOUNDATIONS



Preprocessing + Inverted Indexing

Core mapping of tokens to document IDs, enabling rapid lookup of keyword occurrences.



Lexical Scoring

TF-IDF and BM25 reward document frequency and term importance for exact matches.



Evaluation

Quantitative evaluation methods used to measure retrieval relevance, ranking quality, and overall search effectiveness

TRADITIONAL VS. AI-POWERED SEARCH

Traditional: Keyword Retrieval

Focuses on Information Retrieval (IR) and exact term overlap. Systems are mathematical, explainable, and highly efficient.

TF-IDF / BM25



- Relies on statistical scoring of keywords.
- Explainable: matches are traceable to terms.
- Predictable but limited to literal intent.

AI-Powered: Semantic Synthesis

Uses RAG to understand intent. Systems summarize multiple sources into direct, human-readable answers.

RAG / LLM GEN



- Semantic meaning over exact keywords.
- Summarizes across multiple documents.
- Hallucination risk: harder to verify sources.

CORE RESEARCH QUESTION

How do AI-powered retrieval systems compare to traditional IR systems when evaluated on a shared dataset?



Retrieval Quality

Examining effectiveness and ranking behaviors between traditional lexical and semantic approaches.



User Perception

Analyzing trust, speed, and perceived usefulness of direct answers vs. document lists.



Relevance Delta

Investigating the trade-off between contextual understanding and strict lexical precision.



Strategic Trade-offs

Identifying specific strengths and weaknesses to inform modern retrieval system design.

BEIR NATURAL QUESTIONS (NQ)



Corpus: The Knowledge Base

A massive collection of Wikipedia articles serving as the ground truth. Over 100k+ documents representing the raw information pool for retrieval.



Queries: Real User Intent

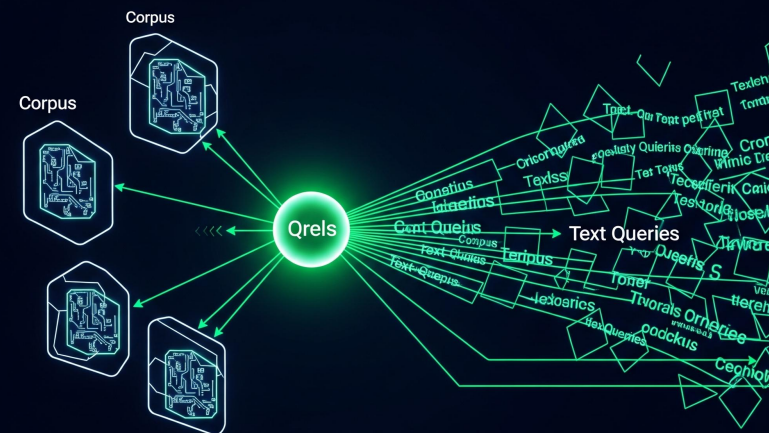
Natural language questions sourced from Google Search. These reflect genuine information-seeking behavior rather than synthetic prompts.



Qrels: Relevance Judgments

Expert-annotated mapping between queries and documents. Defines which documents actually answer a query, enabling objective scoring.

BEIR standardizes NQ for zero-shot evaluation, making it a suitable benchmark for testing how retrieval models generalize to new domains.



THE SEARCH ENGINE ARCHITECTURE



txtai retrieval

Core workflow manager that orchestrates indexing, vector storage, and similarity search for retrieval.



Hugging Face MiniLM

Sentence transformer model generating compact, high-performance dense vector embeddings.



Streamlit Framework

Python-native frontend providing real-time interactive comparison of retrieval results and metrics.



Integration: Modular architecture allows for seamless swapping of transformer models while maintaining UI consistency.

TECHNICAL FOUNDATION

Dataset & Environment

BEIR Natural Questions: Real user-style queries paired with relevance judgments.

Corpus Consistency: Ensured both systems operated on the same shared corpus to isolate performance variables.

Streamlit Interface: Rapid prototyping for interactive comparison of retrieval outputs.

Semantic Engine: txtai embeddings powered by the *MiniLM* sentence transformer.

Semantic Model

- Converts queries/docs into vector embeddings.
- Retrieves conceptually related content.
- Overcomes limitations of exact keyword matching.



THE SEARCH ENGINE INTERFACE

Search Engine - BEIR NQ

Enter your search query:

what type of reproduction do whiptail lizards use

Search

AI Overview

The whiptail lizard reproduces through parthenogenesis.

Key characteristics of whiptail lizard reproduction include:

- The New Mexico whiptail is an all-female species that lays up to four unfertilized eggs in mid summer [2].
- This reproductive strategy allows the population to reproduce and evolve into a unique species capable of reproduction [3].
- Whiptail lizards, like other squamate reptiles, have a ZZ pair of sex determining chromosomes for males and a ZW pair for females [4].

Sources:

[1] [New Mexico whiptail](#)

[2] [Vertebrate](#)

IR Evaluation

Precision@5

0.600

Recall@5

0.750

F1@5

0.667

nDCG@5

0.832

Search Results

1. [New Mexico whiptail](#)

https://en.wikipedia.org/wiki/New_Mexico_whiptail | Score: 0.7802

The New Mexico whiptail lizard is a crossbreed of a western whiptail, which lives in the desert, and the little striped whiptail, which favours grasslands. The lizard is a female-only species that reproduces by producing an egg through parthenogenesis. Despite reproducing asexually, and being an all-female

Interactive Components

- **Streamlit Frontend:** Lightweight, real-time data visualization.
- **AI Summaries:** Context-aware overviews of top results.
- **Technical Metrics:** Real-time display of retrieval performance.
- **Transparency:** Ranked results with links and content snippets.

EXAMPLE QUERY COMPARISON

Query: "define visceral"

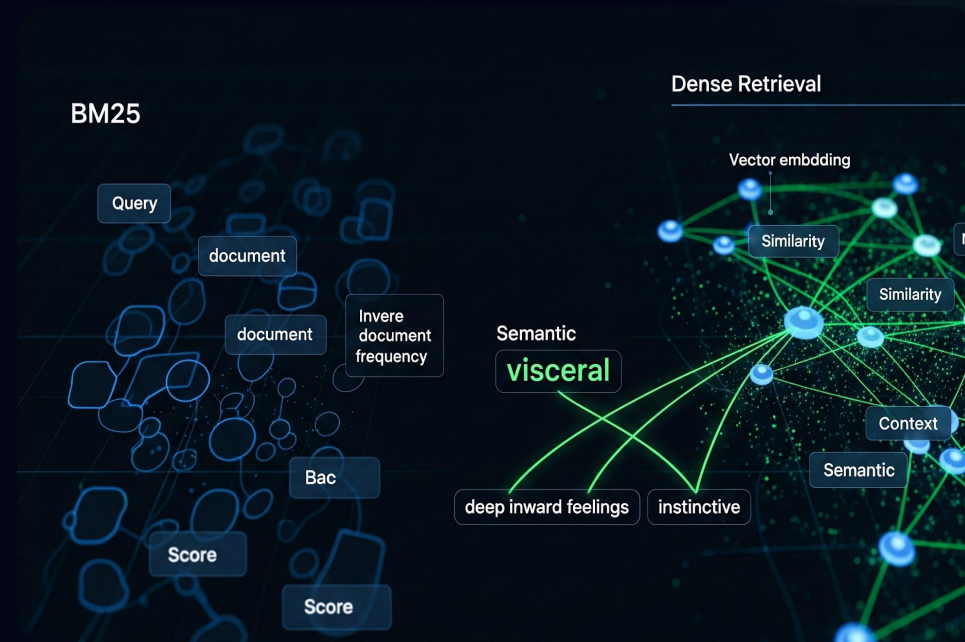
Testing lexical vs. semantic understanding of abstract concepts.

BM25 (Keyword-Based)

- Relies on exact term overlap.
- Struggles with "vocabulary mismatch".
- Returns weakly related or partial matches.

Dense Retrieval (Semantic)

- Operates semantically, not purely lexically.
- Bridges gap between different phrasings.
- Captures conceptual intent and meaning.



Success Factor: Semantic retrieval succeeds by mapping meaning to vector space, overcoming the literal limitations of keyword matching.

EVALUATION METRICS

Precision@k

Measures correctness: how many retrieved results were actually relevant.

Recall@k

Measures coverage: how much of the total relevant info was successfully found.

F1 Score

A balanced metric combining precision and recall to manage performance tradeoffs.

nDCG

Ranking quality: penalizes relevant results that appear lower in the list.



"Evaluating not just if information was retrieved, but how effectively it was ranked for the user."

Metric

Precision

Recall

F1 Score

nDCG

What it Measures

Correctness

Coverage

Harmonic Mean

Ranking Quality

CHALLENGES & METHODOLOGY ADAPTATION



IRB & Participant Access

Original plan for external participants restricted by IRB timing; pivoted to researcher-led and LLM-evaluations.



Hallucination Control

Mitigated risk of unsupported info by strictly constraining generation using retrieved context documents.



Retrieval Anomalies

Encountered retrieval inconsistencies and duplicate titles in the corpus, affecting diversity metrics.



Substitute Evaluation

Adopted a hybrid evaluation process involving partner researchers to maintain exploration of system behavior.



"Research rarely goes perfectly; adapting the methodology was a core part of the learning process."

THE RELEVANCE GAP

"The Good Doctor" Case

Query: Who plays the doctor on The Good Doctor?

Issue: Semantic retrieval found docs about the *TV Show*, but the *Benchmark* only marked the doc mentioning *Freddie Highmore* as relevant.

Findings: Benchmarks often fail to capture "useful" context that human users value, leading to misleadingly low metric scores.



| EXPERIMENTAL RESULTS & ANALYSIS



WHY WE DESIGNED THIS SURVEY

Each question maps to a **core evaluation dimension** of user search behavior:

Search Preference

Investigating behavioral choice between systems.

Perceived Speed

Efficiency perception vs. technical reality.

Task-Based Performance

Analyzing context dependency in search.

Trust & Transparency

Understanding why users select specific systems.

Goal: Understand how users evaluate systems beyond simple surface-level preferences.

| SURVEY QUESTIONS & MEASUREMENT

Topics	Dimension Measured	Metric Utility & Logic
1	System Preference	Establishes baseline behavior comparing AI vs. traditional mental models. Captures the default user approach to search systems.
2	Perceived Speed	Separates technical "real speed" from user "felt speed" and efficiency perception. Investigates immediate synthesis vs. processing time.
3	Quick Queries	Tests for context-dependent usage and adaptive behavioral shifts. Measures if preference changes based on the specific nature of the query.
4	Trust & Reliability	Assesses user confidence in information accuracy and system dependability. Evaluates the perceived authority of results generated by AI vs. traditional indexes.

We structured the survey to test whether search preference is static or context-dependent.

STUDY CONSTRAINTS & COLLECTION

Methodological Constraints

- IRB approval timeline constraints prevented formal university-wide student recruitment.
- Unable to formally survey public participants during the active study window.
- Methodology adapted to exploratory framework to maintain study momentum.
- Findings viewed as **exploratory** rather than fully generalizable.

100

Researcher Responses

Internal baseline metrics generated by the core research team for comparison.

50

AI Responses

Simulated perspectives across various student backgrounds and search expertise.

Simulated AI Agents Strategy

Models were prompted to simulate a diverse range of users to broaden perspective while maintaining controlled conditions. This allowed for analyzing patterns in trust, efficiency, and system preference across different "user" personas.

Note: Findings focus on identifying emerging system behavioral patterns rather than making population-level generalizations.

SURVEY METHODOLOGY & DESIGN

Comparative Framework Design

- Designed a survey to measure user preference between traditional and AI-enhanced search systems.
- Maintained high **structural consistency** by using identical questions across all response groups.
- Comparison Groups: Human Researchers (Shahab and Sarang) and Simulated AI Responses.
- Evaluated how factors like **trust, familiarity, and task type** influence preference shifts.

Core Evaluation Pillars

Overall Preference

Default choice between systems

Perceived Speed

Efficiency vs. technical reality

Quick Queries

Behavior during everyday searches

Trust & Reliability

Confidence in information accuracy

Goal: Identify broader behavioral trends and context-dependent preference patterns.

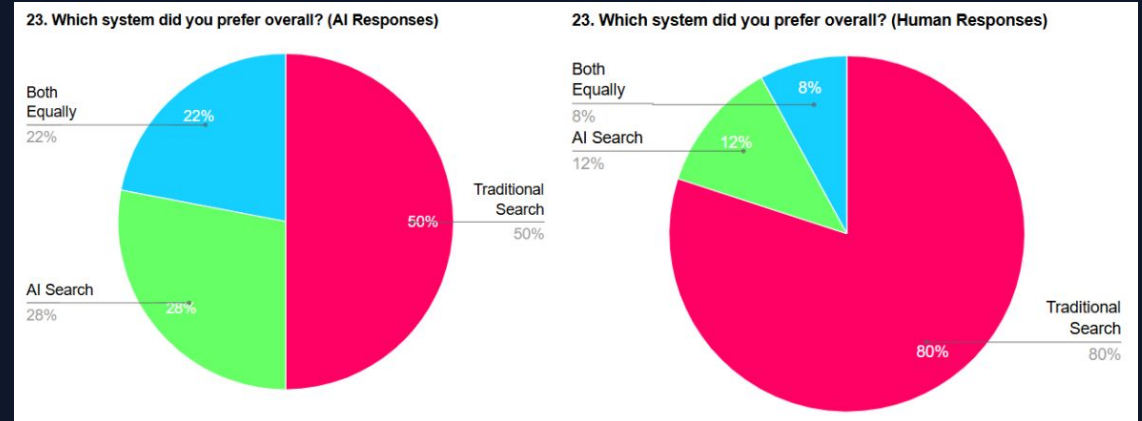
What the Responses Reveal: User Behavior

Traditional Dominance & Verification

Traditional Search remains the baseline for most users due to established trust and direct source access.

- **80% of respondents** prefer traditional systems for complex verification needs.
- Synthesized answers are secondary to **manual document control**.
- Trust is tied to familiarity with existing search interfaces.

Comparative Overall Preference: Human vs. AI Responses



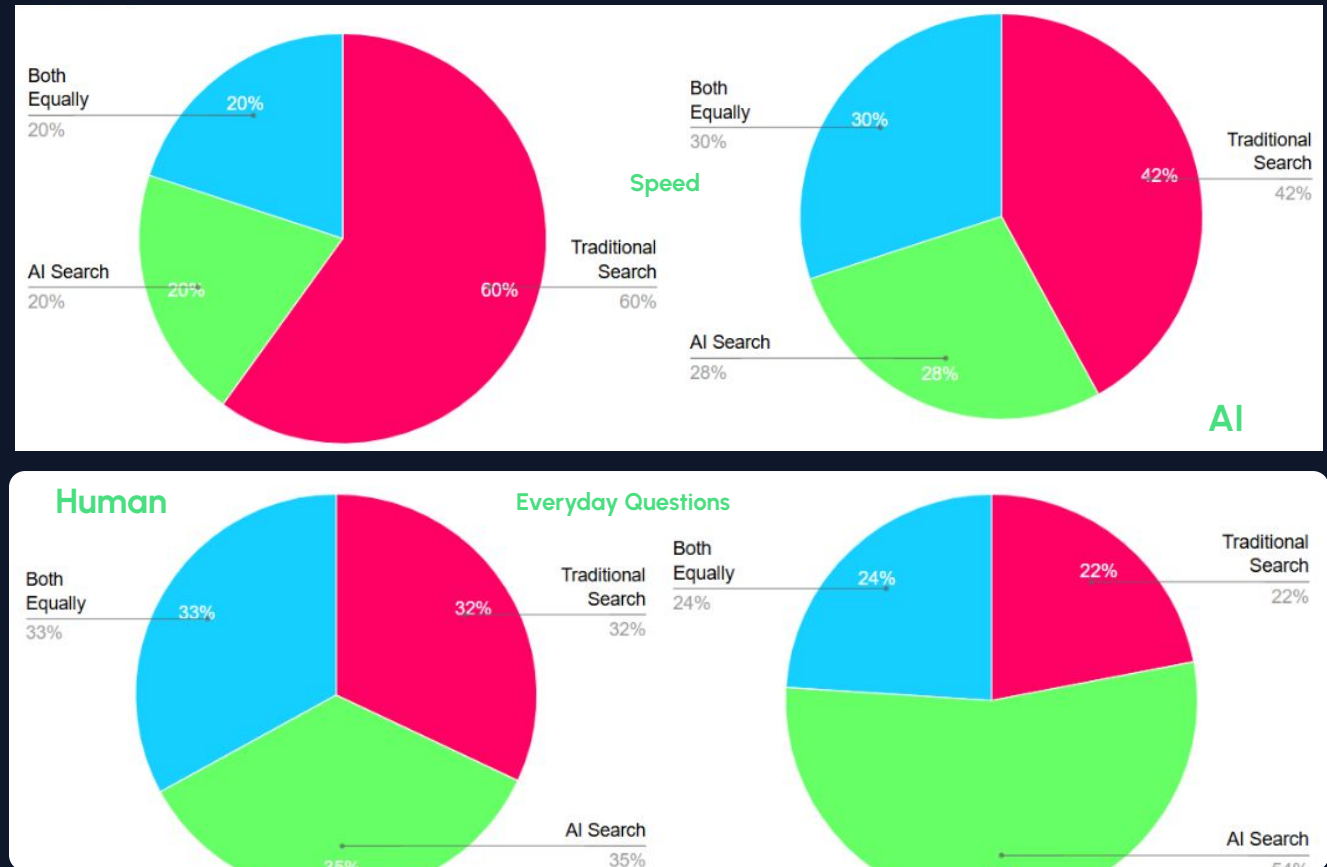
"The results suggest search behavior is not uniform, but situational."

Speed & Convenience: Query-Specific Preferences

Efficiency vs. Everyday Search

Traditional search excels in verification, and also dominates when speed is the primary user objective.

- **Speed Perception:** Users perceive traditional responses as significantly faster for finding initial information.
- **Quick Queries:** For everyday searches, preference shifts towards **AI-synthesized** summaries.
- **Efficiency Gap:** "Time to Answer" favors AI for conceptual understanding.



"User preference is driven by the trade-off between speed and source-level certainty."

Discussion: Quantitative Results Summary

Retrieval Performance Analysis

The traditional system demonstrates **overall strong and consistent retrieval performance** across all key evaluations.

- **Balanced Reliability:** Precision and recall are closely aligned, indicating accurate results without sacrificing coverage.
- **Superior Ranking:** The high nDCG score confirms that the most relevant documents are effectively prioritize at the top.
- **Uniform Metrics:** Metrics remain stable across multiple dimensions, suggesting a robust retrieval pipeline.

Average Recall

0.7991

Average Precision

0.8046

Average F1 Score

0.7941

Average nDCG

0.8460

**STABLE RETRIEVAL QUALITY ACROSS ALL EVALUATION
DIMENSIONS**

"Users evaluate systems through a perceptual lens, while benchmarks evaluate them through a structural one."

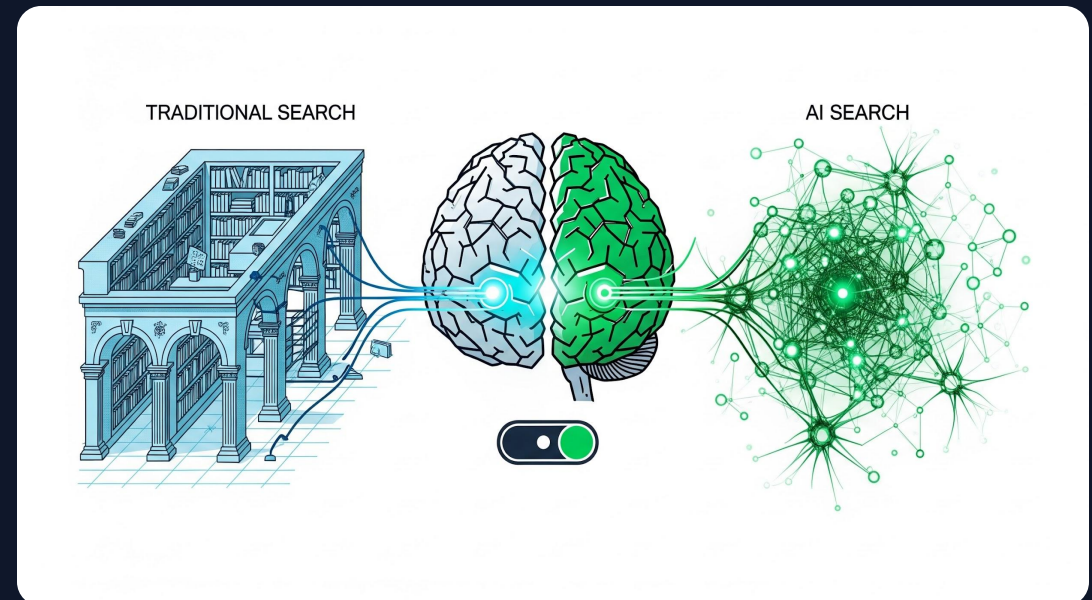
Key Insight: Search Preference is Context-Dependent

Strategy Based on Task Type

Users do not choose a search system universally; they select a **strategy adapted to user intent** and situational needs.

- **Traditional Dominance:** Remains the baseline for complex verification, manual control, and source-level transparency.
- **AI Competitiveness:** Becomes the preferred choice for quick, everyday queries where convenience outweighs source inspection.
- **Perceived Efficiency:** AI is often experienced as "faster" despite internal latency, as it delivers direct synthesized responses.

DYNAMIC ADAPTATION TO USER INTENT



"This suggests future search systems should adapt dynamically to user intent rather than assume a single optimal interface."

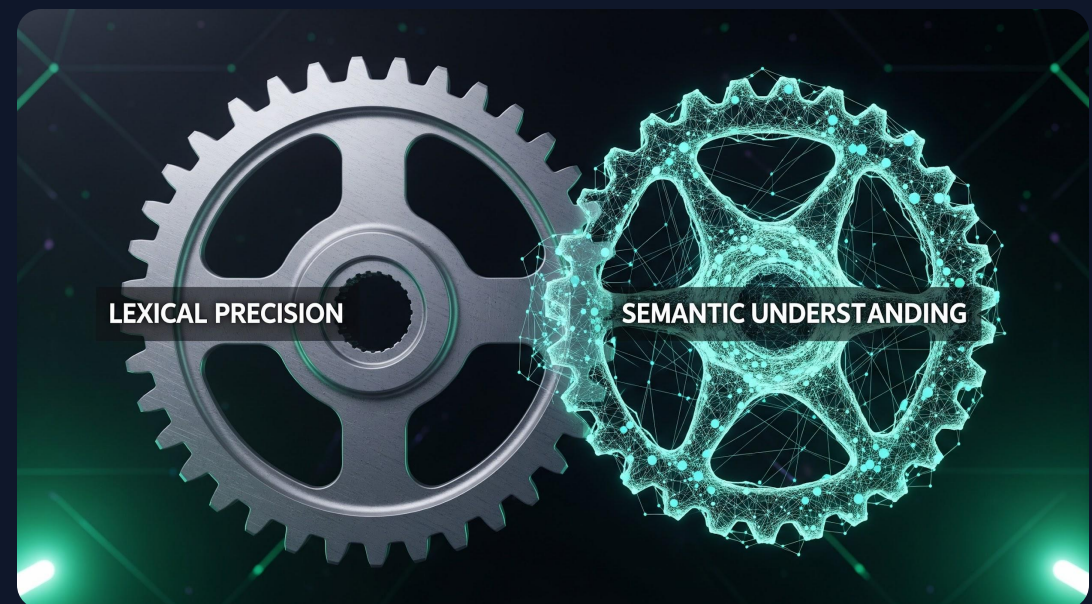
Discussion: Core Retrieval Tradeoffs

System Dichotomy

Approaches offer unique strengths based on task requirements.

- **Lexical vs. Semantic:** Keyword precision vs. conceptual meaning.
- **Explainability:** Statistical transparency vs. abstract AI reasoning.
- **Efficiency:** Lightweight methods vs. high-latency LLM power.
- **Utility:** Semantic understanding doesn't always equal relevance.

BALANCING PRECISION WITH CONCEPTUAL DEPTH



"Stronger semantic understanding does not always guarantee better practical relevance."

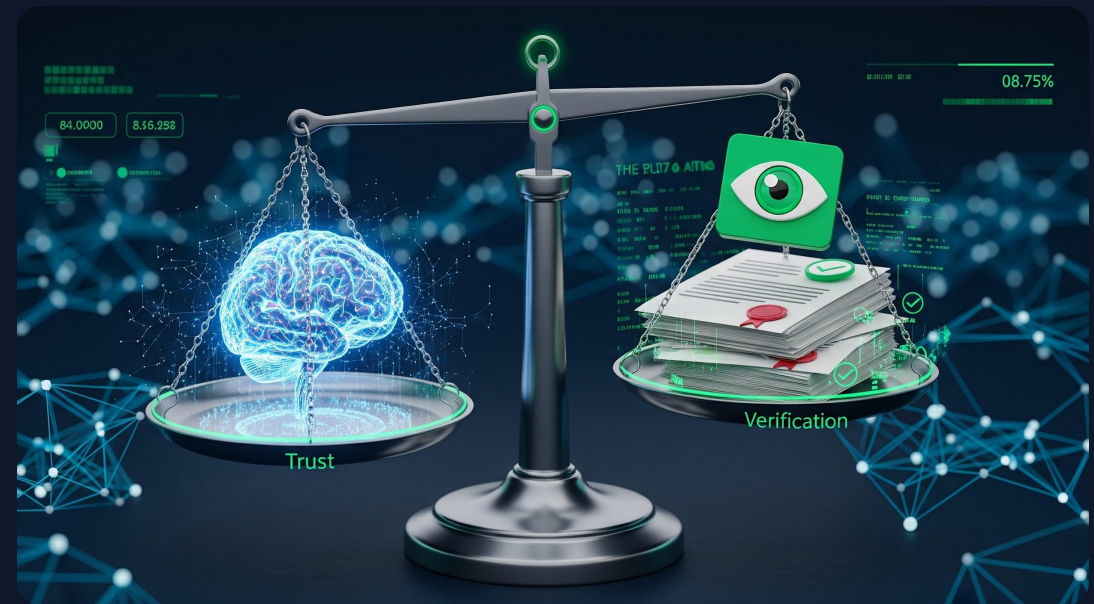
Discussion: Ethical & Practical Considerations

Integrity & Accountability

Generative synthesis introduces critical risks to data veracity and behavior.

- **Hallucinations:** False but confident answers in high-stakes tasks.
- **Transparency Deficit:** AI can obscure reasoning and source weighting.
- **Over-Reliance:** Users may trust fluent phrasing without verification.

NAVIGATING THE TRUST-TRANSPARENCY PARADOX



"Maintaining transparency and encouraging source verification will remain critical for ethical AI adoption."

Limitations & Scope

Study Constraints

Evaluated via a proof-of-concept framework with procedural boundaries.

Limited Participants: Small scale due to time and IRB constraints.

Synthetic Baseline: Relied on AI/researcher responses over human data.

Design Focus: Controlled, partially synthetic dataset reduced response variability, allowing clearer isolation of system-level behavior rather than population diversity.



"Exploration of retrieval behavior across contexts, not universal proof."

Future Research Directions

Scaling & Optimization

Expanding the project via larger datasets and advanced retrieval architectures.

- **User Validation:** Surveys with real users to strengthen statistical validity.
- **Hybrid Architectures:** Combining lexical ranking with semantic systems.
- **LTR Integration:** Dynamic machine learning to optimize rank signals.
- **Advanced LLMs:** Evaluating next-gen models for better comprehension.



EXPANDING SYSTEM CAPABILITIES

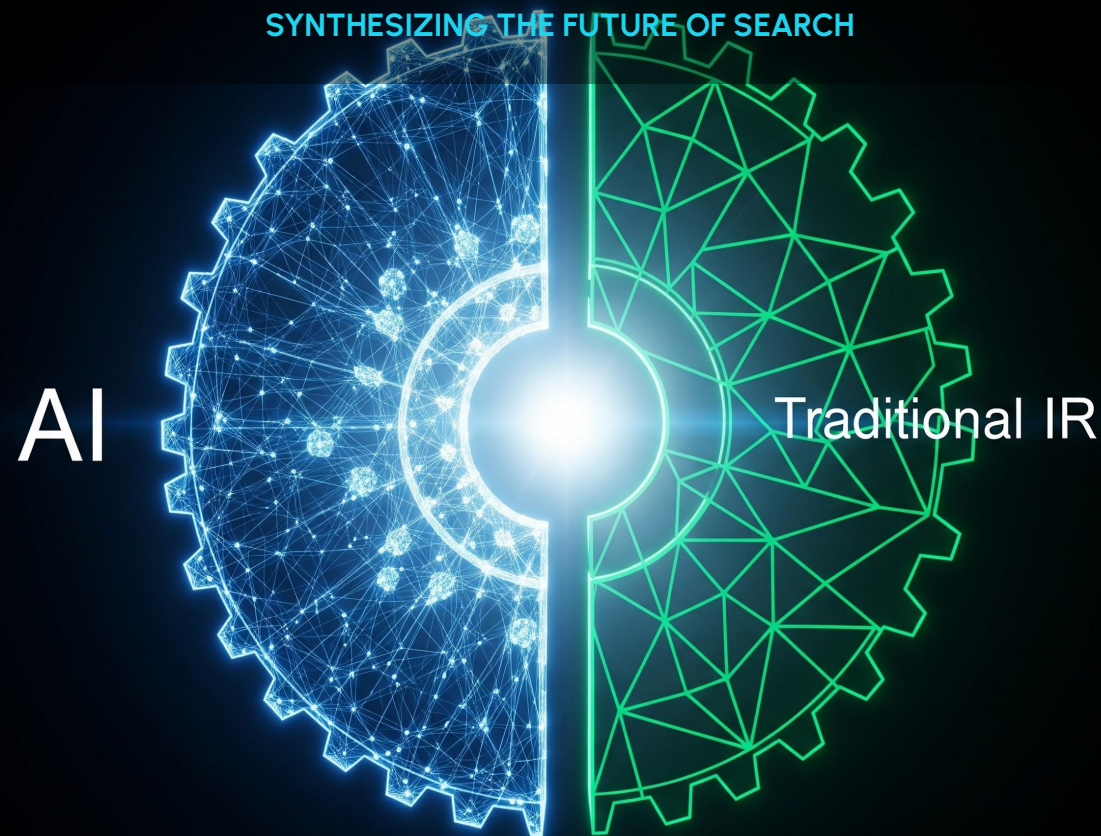
"Advancing retrieval and user-behavior modeling through larger-scale integration."

Final Conclusions

Key Takeaways

Synthesizing technical evaluation and behavioral analysis outcomes.

- **Semantic Mastery:** AI retrieval excels at handling vocabulary mismatch and natural language context.
- **Traditional Strength:** Keyword-based IR remains essential for precision, speed, and transparency.
- **Hybrid Future:** Optimal performance lies in the combination of semantic convenience and verified foundations due to **context**.



"The future of search is not purely traditional or generative, but a combination of both working together."

What We Learned

Reflection

Key insights from the engineering and research journey.

- **Research:** Adapting to changing methodologies and challenges.
- **Systems:** Practical experience with LLMs and embeddings.
- **Evaluation:** Decoding ranking behavior and user perception.
- **Theory:** Bridging IR theory with real-world datasets.



"This project taught us much more than just how search systems work technically."



Thank You

Questions & Discussion

Honors Thesis Defense